Supporting Information for

Emerging β-sheet rich conformations in super-compact Huntingtin exon-1 mutant structures

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Side chain-side chain hydrogen bonds in interior glutamines

Protein solubility depends on the side chain-water hydrogen bonding propensity, which will be directly affected by whether amino acid side chains are on surface or buried inside of the protein. Thus, we calculated the average number of side chain-side chain hydrogen bonds in the polyQ for buried glutamines and for all glutamine residues. Here, interior glutamines are defined as glutamines having less than 4 water molecules within 3Å. We found that the number of side chain-side chain hydrogen bonds of buried glutamines in polyQ is even greater than that of normal GLN in structured proteins in PDB (**Fig S1**).

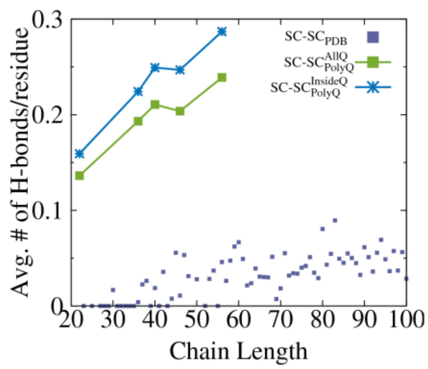

Figure S1 Average number of side chain-side chain (SC-SC) hydrogen bonds between interior glutamines in polyQ, defined as glutamine residues having less than four water molecules within 3Å; all glutamines in polyQ; and proteins from the PDB. The overall polyQ region (green square) has more side chain-side chain hydrogen bonds than general structured proteins (violet square). The buried glutamines (blue star) on average have even more SC-SC hydrogen bonds.

Average number of hydrogen bonds in surface Glutamines

We calculated the average number of side chain-water and side chain-protein hydrogen bonds for surface GLN residues in each of our htt variants and in globular proteins from the PDB. Surface GLN residues were defined as residues having at least 6 water molecules within 3.0 Å of the residue. We found that the propensity for surface GLN residues to form side chain-water hydrogen bonds was lower in polyQ than in globular protein. The propensity to form side chain-protein hydrogen bonds was roughly the same between polyQ and globular proteins.

	Q22	**Q36**	**Q40**	**Q46**	**Q56**	**PDB**
Side chain-Water	1.6	1.6	1.6	1.6	1.6	2.9
Side chain-Protein	0.12	0.13	0.14	0.15	0.16	0.16

Table S1. Average number of side chain-water and side chain-protein hydrogen bonds for surface glutamines in polyQ and in globular proteins.

Average number of interdomain hydrogen bonds

To understand how interactions between the different htt exon-1 changes with Q-length, we calculated average number of hydrogen bonds between N17 and polyQ, C38 and polyQ (Fig. S2A), and N17 and C38 (Fig. S2B). This analysis showed that the average number of hydrogen bonds between polyQ and the flanking domains increases as a function of Q-length. The number of hydrogen bonds between N17 and C38, however, increases from Q22 to Q36 and then decreases with increasing Q-length.

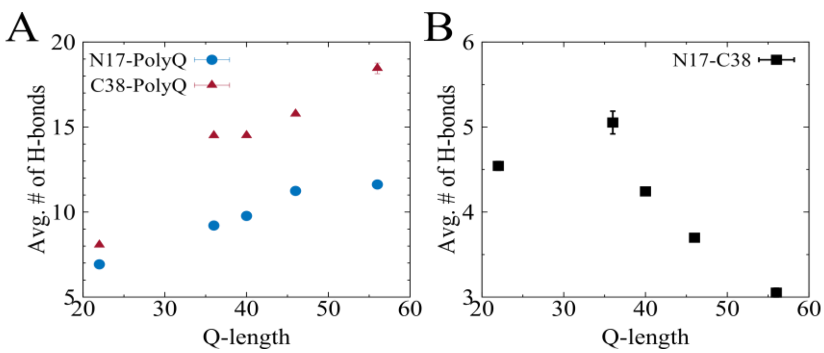

Figure S2. Average number of interdomain hydrogen bonds for N17-PolyQ, C38-PolyQ (A) and N17-C38 (B). Hydrogen bonds related to PolyQ increases with increasing Q-length (A) whereas the number of N17-C38 hydrogen bonds decreases. For more details, refer main text

Correlation of interdomain hydrogen bonds with secondary structures

In order to investigate how the secondary structure of the polyQ region affects N17-C38 contacts, we calculated 2D distributions of α- or β-content in the polyQ region vs. the number of hydrogen bonds between N17 and C38 and then fit them using a linear regression model (**Fig. S3**). The number of β-sheet contents shows some negative correlation with N17-C38 contacts while α-helical content show weak correlation.

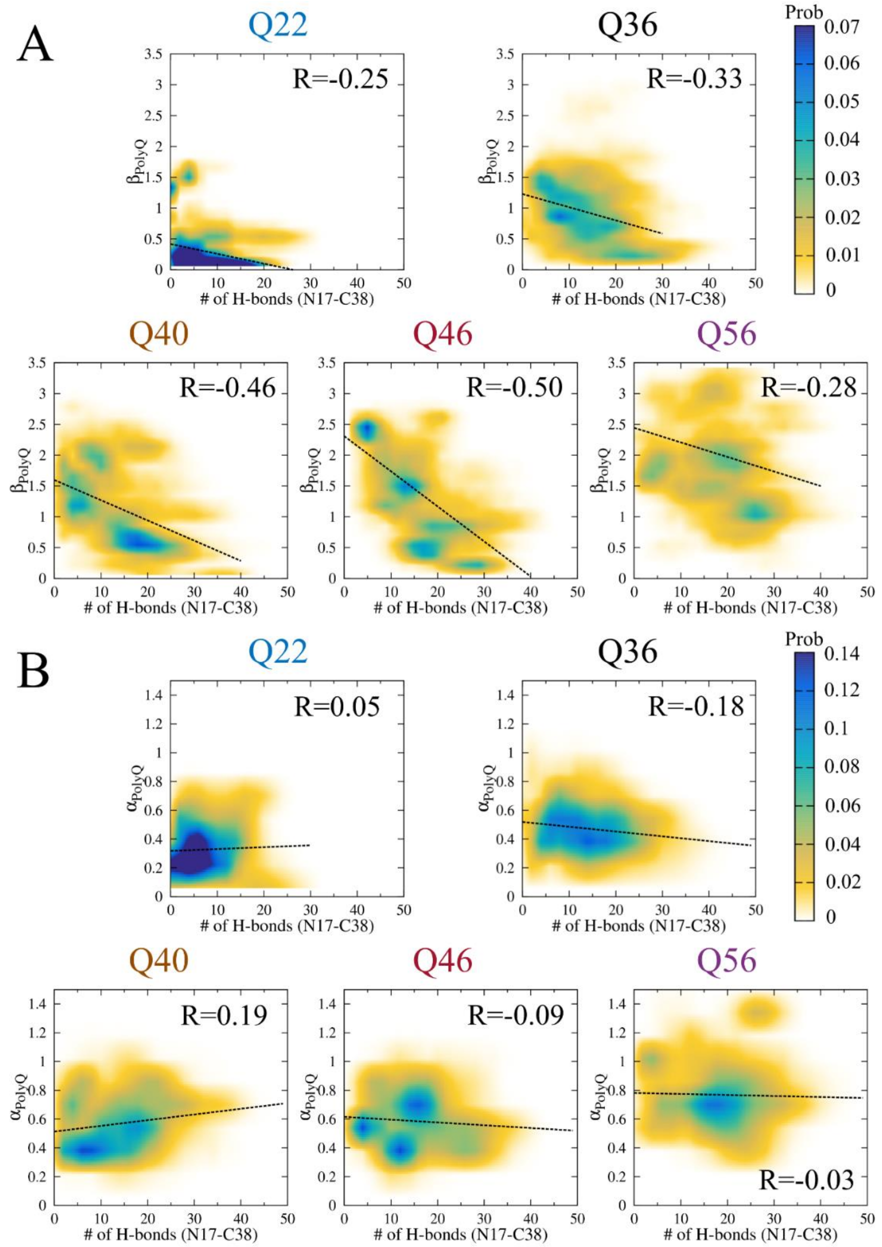

Figure S3. The correlation between secondary structure scores and hydrogen bond counts between N17 and C38 suggests that the formation of β-sheet rich conformations decreases the interaction between the N17 and C38 domains. (A) 2D histograms of the number of hydrogen bonds between N17 and C38 regions and β-sheet scores for polyQ regions. The dashed line shows the fit determined with linear regression. The correlation coefficient, R, is shown in each plot. The 2D histograms and linear regression models show that there is negative correlation between β-sheet structures and N17-C38 contacts. (B) 2D histograms of the number of hydrogen bonds between N17 and C38 regions and α helix scores for polyQ regions. In contrast to β-sheet content, there does not seem to be a correlation between α helix content and N17-C38 interaction.

Convergence of the simulation results

To ensure the convergence of our simulation results, we examined autocorrelations functions for the radius of gyration, the end-to-end distance, the fractions of α-helical or β-sheet content, and the solvent accessible surface area (SASA). Most of autocorrelation functions rapidly decayed to 0.1 within 10 ns, which is shown for Q56 in **Fig. S4**. Only SASA converged at a slower rate, but the autocorrelation function still vanished within 100 ns (Shown for Q56 in insets of **Fig. S4**). **Figure S5** also shows the distributions of SASA with respect to time windows, which confirms the same convergence. Thus, we concluded that all structural biases originating from the initial structures dissipated within 100 ns, and as a result, we only analyzed the data in the remaining 400 ns.

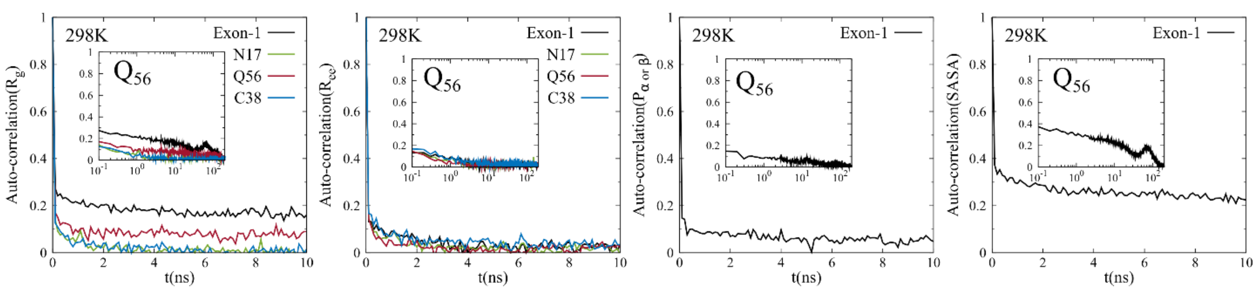

Figure S4. Autocorrelations of various quantities for Htt exon-1 a 56-mer polyQ region (Q_{56}) on linear (main) and log-log scales (inset). R_g is the radius of gyration, R_{ee} is the end-to-end distance, $P_{\alpha\ or\ \beta}$ is the percentage of α-helical or β-sheet content, and SASA is the solvent accessible surface area in a snapshot. All autocorrelations vanish within 100ns, indicating loss of memory from initial conformation at 100ns.

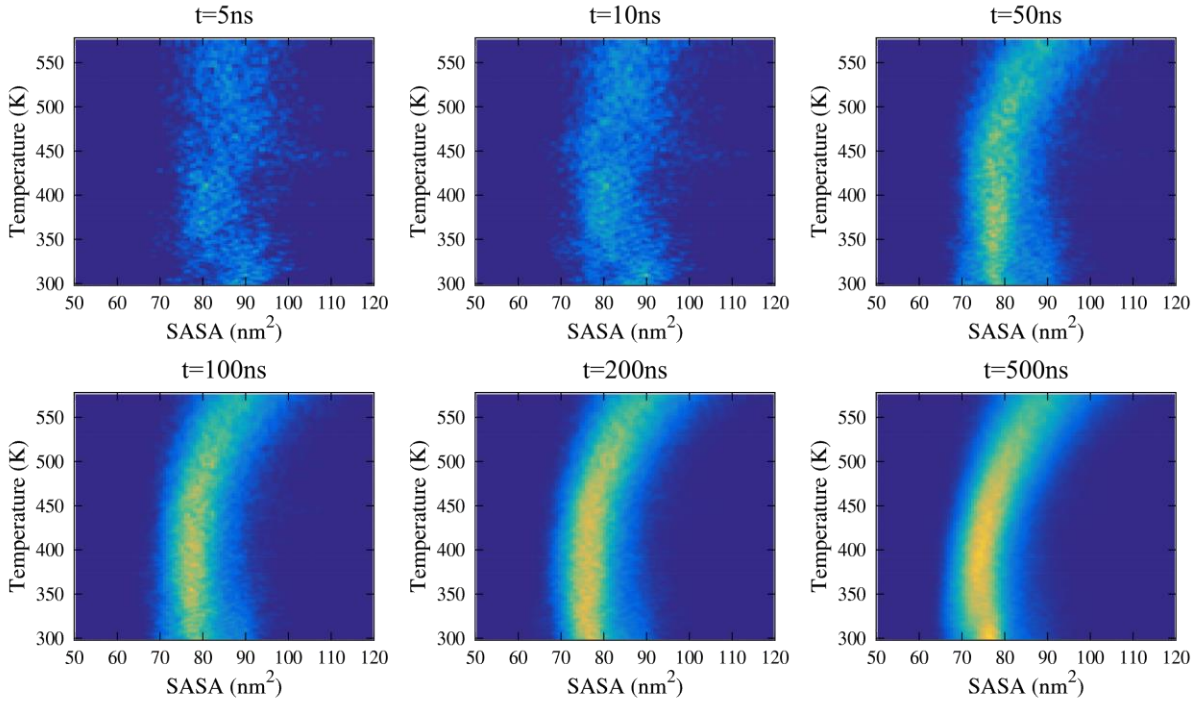

Figure S5. The changes in SASA distributions with respect to time windows for collecting samples. The left-top panel is a 2D distribution for the first 5ns data and right-bottom panel is for 500ns. The plot visually confirms that there is no qualitative difference in first 200ns and 500ns results, indicating our simulations are converged at 500ns.